

# **Statistics Portfolio**

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# 1 Statistics Portfolio

## 1.1 About

Valdinar Pires Student at the University of Campinas (UNICAMP), with an interest in statistical analysis, data visualization, and statistical modeling.

This portfolio brings together projects developed throughout my academic training, with an emphasis on applying statistical concepts to real-world and historical datasets.

## 1.2 Projects

The projects presented here explore different statistical methods and approaches, combining theoretical foundations with practical analysis in R and other data-related tools.

### 1.2.1 Current projects

- **Semmelweis Handwashing Study** Analysis of historical data from the Vienna General Hospital, investigating the relationship between handwashing and maternal mortality.

More projects will be added as my academic portfolio develops.

## 1.3 Tools

The main tools used across the projects include:

- R
- Quarto
- Git and GitHub

Additional tools and programming languages will be incorporated as the portfolio expands.

## 2 About Me

I am a Statistics student at the University of Campinas (UNICAMP), interested in understanding how statistical methods can be used to extract information from data and support quantitative reasoning.

My academic training combines statistical theory, mathematical foundations, and practical data analysis. Through coursework and independent projects, I have been developing skills in statistical programming, data visualization, modeling, and statistical inference.

### 2.1 Academic Interests

My main interests include:

- Statistical inference
- Sampling and survey methodology
- Regression and statistical modeling
- Experimental design
- Exploratory and descriptive data analysis
- Data visualization

### 2.2 Working with Data

I am particularly interested in the process of moving from raw data to meaningful statistical conclusions: understanding the structure of a dataset, selecting appropriate methods, evaluating assumptions, and communicating results clearly.

This portfolio serves as a record of that development, bringing together projects completed during my academic training and documenting the methods used in each analysis.

## 3 Projects

This section presents selected statistical projects developed throughout my academic training.

Each project focuses on a specific dataset, statistical question, or methodological problem, combining statistical reasoning with reproducible data analysis.

### 3.1 Semmelweis Handwashing Study

#### Statistical analysis of handwashing and maternal mortality

This project analyzes historical data from the Vienna General Hospital between 1841 and 1846 to investigate the relationship between handwashing practices and maternal mortality.

The analysis includes:

- Comparison of mortality between two clinics;
- Descriptive statistics and data visualization;
- Comparison of mortality before and after the introduction of handwashing;
- Comparison of mean, median, and mode;
- Hypothesis testing.

**Tools:** R · tidyverse · ggplot2 · Quarto

[View project →](#)

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### 3.2 More Projects

Additional projects will be added as my academic portfolio develops, covering topics such as sampling, regression, experimental design, probability, statistical inference, and data analysis.

## 4 Language R

Assignment

### 4.1 Introdução

The Hungarian physician [Dr. Ignaz Semmelweis](#) was born in 1818 and worked at the Vienna General Hospital. His work on puerperal fever was of great importance. Puerperal fever is a potentially fatal disease that affects women after childbirth. In the early 1840s, around 10% of women who had given birth at the Vienna General Hospital died from this disease. Dr. Semmelweis discovered the cause of puerperal fever: contaminated hands of doctors who attended women during childbirth. However, his discovery was not accepted for many years.

### 4.2 Observing the situation

Let's reanalyze the data that led Dr. Semmelweis to discover the importance of handwashing.

```
# Loading the tidyverse package  
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.4.3

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'tibble' was built under R version 4.4.3

Warning: package 'tidyr' was built under R version 4.4.3

Warning: package 'readr' was built under R version 4.4.3

Warning: package 'purrr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'stringr' was built under R version 4.4.3

Warning: package 'forcats' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.3

```
# Reading the dataset about annual deaths and saving it in the data_death object
data_death <- read_csv("mortes_anuais_clinica.csv")
```

```
# Renaming the columns
data_death <- data_death %>%
  rename(
    year = ano,
    births = nascimentos,
    death = mortes,
    clinical = clinica
  )
```

```
#Renaming some values
data_death <- data_death %>%
  mutate(
    clinical = recode(
      clinical,
      "clinica 1" = "clinic 1",
      "clinica 2" = "Clinic 2"
    )
  )
```

```
# Printing the dataset
data_death
```

```
# A tibble: 12 x 4
  year births death clinical
<dbl> <dbl> <dbl> <chr>
1  1841   3036   237 clinic 1
2  1842   3287   518 clinic 1
3  1843   3060   274 clinic 1
4  1844   3157   260 clinic 1
```

|    |      |      |     |        |   |
|----|------|------|-----|--------|---|
| 5  | 1845 | 3492 | 241 | clinic | 1 |
| 6  | 1846 | 4010 | 459 | clinic | 1 |
| 7  | 1841 | 2442 | 86  | Clinic | 2 |
| 8  | 1842 | 2659 | 202 | Clinic | 2 |
| 9  | 1843 | 2739 | 164 | Clinic | 2 |
| 10 | 1844 | 2956 | 68  | Clinic | 2 |
| 11 | 1845 | 3241 | 66  | Clinic | 2 |
| 12 | 1846 | 3754 | 105 | Clinic | 2 |

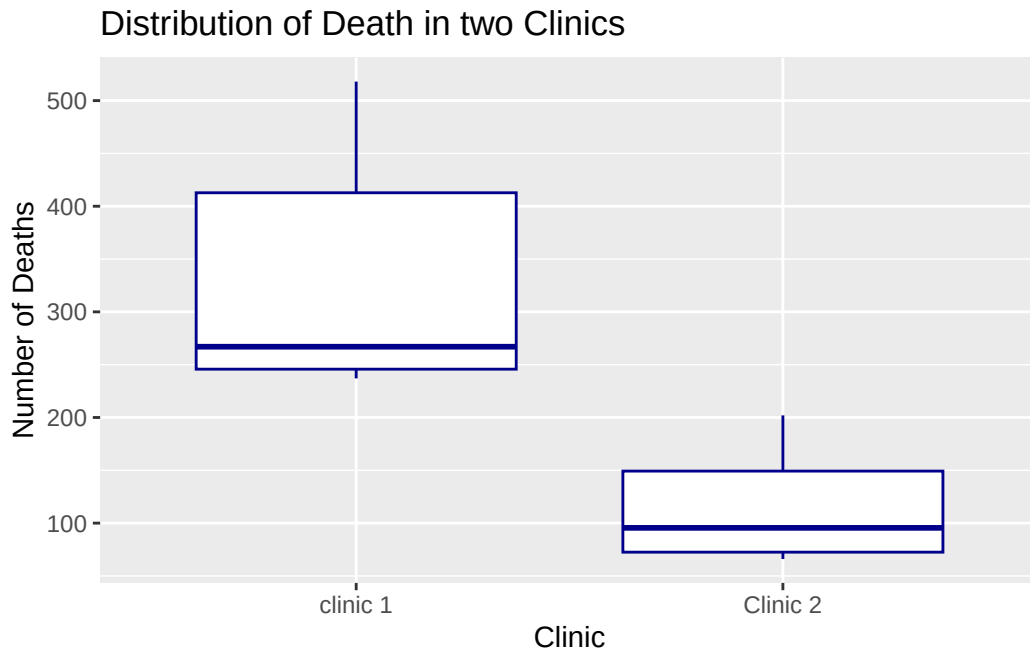
The dataset `data_death` shows the number of childbirths in two clinics at the Vienna General Hospital between 1841 and 1846. We can observe a high number of women who died during childbirth, many of them due to puerperal fever.

To better understand the situation, we will add a new column to the dataset with the proportion of deaths per year in each clinic.

```
data_death <- data_death %>%
  mutate(proportion=death*100/births)
```

Compare a situação entre as duas clínicas usando estatísticas sumárias e/ou gráficos que achar relevantes. Explique exatamente o que você está comparando (qual a variável utilizada?).

```
ggplot(data_death, aes(y=death, x=clinical)) +
  geom_boxplot(colour = "darkblue") +
  labs(
    title = "Distribution of Death in two Clinics",
    x = "Clinic",
    y = "Number of Deaths" )
```



As observed in the graph, there is a difference between the medians of Clinic 1 and Clinic 2. Clinic 1 exhibits a higher median than Clinic 2, which suggests that Clinic 1 records a greater number of deaths. Furthermore, the minimum number of deaths in Clinic 1 exceeds the maximum value observed in Clinic 2, providing additional evidence that Clinic 1 has a higher mortality rate than Clinic 2.

### 4.3 Buscando por uma explicação e propondo uma intervenção

Dr. Semmelweis observed a pattern in these data and became concerned. The only difference between the two clinics was that, in Clinic 1, medical students attended the deliveries, whereas in Clinic 2 midwives attended the women. While the midwives worked exclusively with childbirth, the students also performed autopsies.

Semmelweis began to suspect that something in the autopsy room was influencing the mortality rate. His primary hypothesis was that some form of contaminating agent was being transmitted on the hands of the students, causing puerperal fever. In an attempt to reduce the high mortality rate, he instructed all staff to wash their hands. At the time, however, this measure was highly controversial, since the scientific knowledge of the period did not yet include an understanding of bacteria.

```
# Reading the monthly deaths data for Clinic 1
month <- read_csv("mortes_mensais.csv")
```

```
# Renaming the columns
month <- month %>%
  rename(
    date = data,
    births = nascimentos,
    death = mortes
  )

# Adding the percentage of deaths relative to births
month <- month %>% mutate(proportion = death * 100 / births)

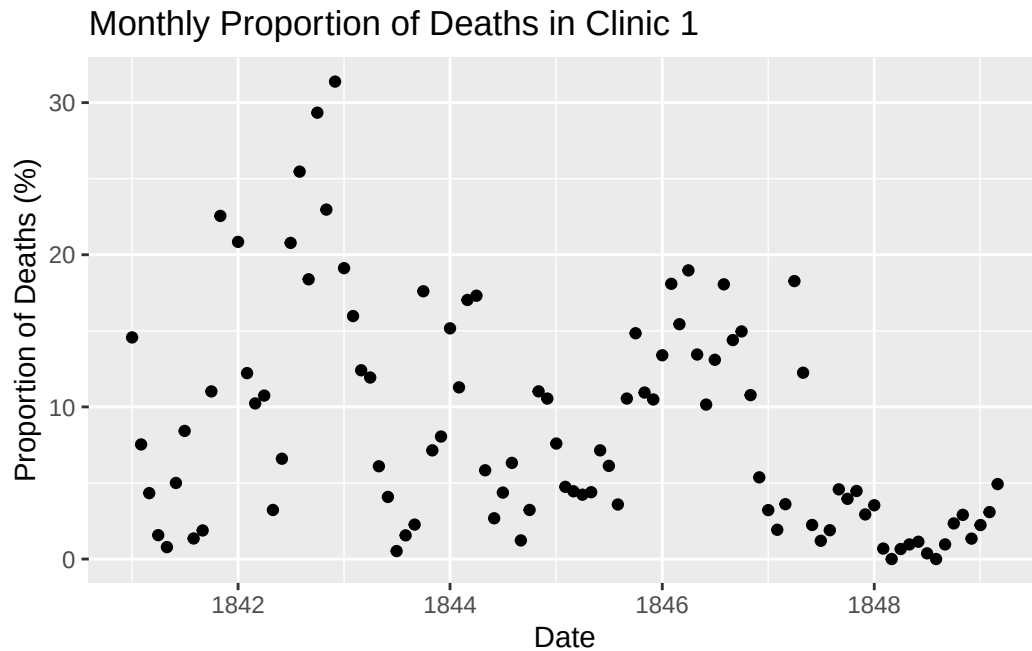
# Printing the first observations of the data set
head(month)
```

```
# A tibble: 6 x 4
  date      births death proportion
<date>    <dbl> <dbl>    <dbl>
1 1841-01-01   254    37    14.6
2 1841-02-01   239    18     7.53
3 1841-03-01   277    12     4.33
4 1841-04-01   255     4     1.57
5 1841-05-01   255     2     0.784
6 1841-06-01   200    10     5
```

## 4.4 The Effect of Handwashing

The proportion of deaths over time is analysed through the following plot:

```
ggplot(month, aes(x = date, y = proportion)) +
  geom_point() +
  labs(
    title = "Monthly Proportion of Deaths in Clinic 1",
    x = "Date",
    y = "Proportion of Deaths (%)"
  )
```



The plot provides an indication of when the order to wash hands was implemented. The change occurred in the summer of 1847.

```
# Date of the start of the order Handwashing  
start_handwashing = as.Date('1847-06-01')
```

```
# Creating a binary column indicating whether the date falls on or after  
# the handwashing order (FALSE = before the order, TRUE = on or after)
```

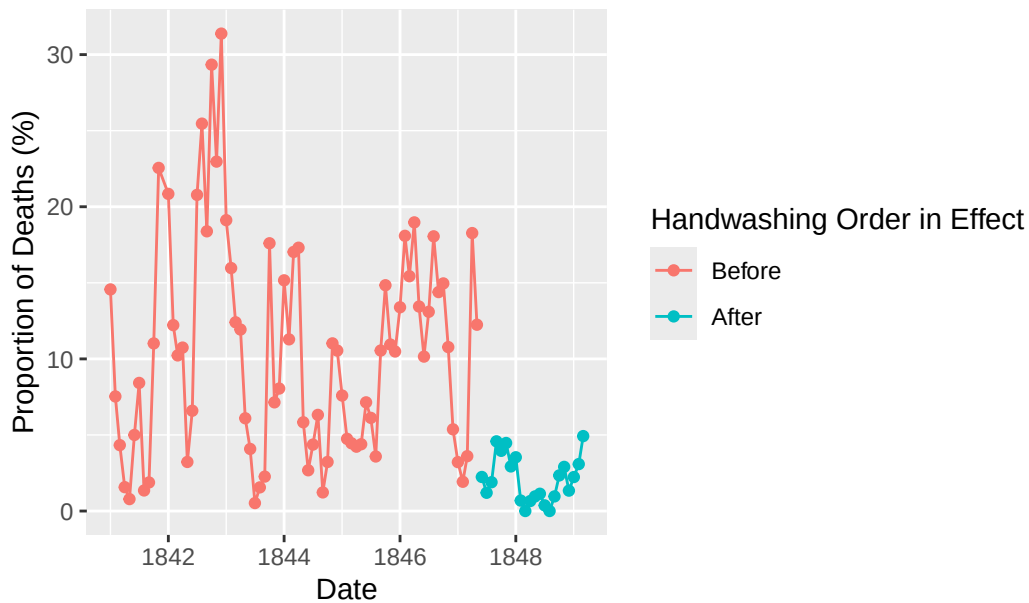
```
month <- month %>%  
  mutate(order_handwash = date >= start_handwashing)
```

## 5 Let's plot a new line graph to better compare the monthly proportion of the data before and after the handwashing order

```
#Line Plot

ggplot(month, aes(x = date, y = proportion, group = order_handwash, colour = order_handwash)) +
  geom_line() +
  geom_point() +
  # Changing the labels in the legend
  scale_color_discrete(labels = c("FALSE" = "Before", "TRUE" = "After")) +
  labs(
    title = "Monthly Proportion of Deaths in Clinic 1",
    x = "Date",
    y = "Proportion of Deaths (%)",
    colour = "Handwashing Order in Effect"
  )
```

## Monthly Proportion of Deaths in Clinic 1



```
# Number of months in which the handwashing order was not yet in effect
count_months_no_order <- function(x) {
  sum(x == FALSE)
}
months_no_order <- count_months_no_order(month$order_handwash)
```

```
# Mean proportion of deaths in the period before the handwashing order
mean_proportion_death_no_handwash <- mean(month$proportion[month$order_handwash == FALSE])
mean_proportion_death_no_handwash
```

```
[1] 10.505
```

```
# Number of months in which the handwashing order was in effect
months_yes_order <- as.numeric(sum(month$order_handwash))
months_yes_order
```

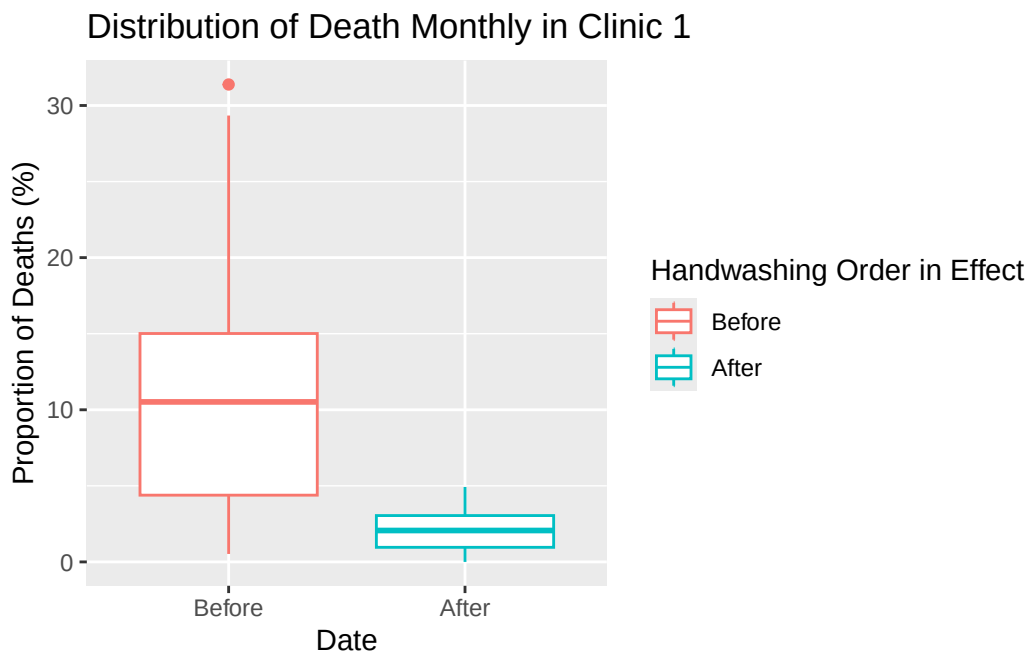
```
[1] 22
```

```
# Mean proportion of deaths in the period after the handwashing order
mean_proportion_death_yes_handwash <- mean(month$proportion[month$order_handwash == TRUE])
mean_proportion_death_yes_handwash
```

[1] 2.109338

Comparing the situation, focusing on the proportion of death between wahs or not wahse order. Compare a situação, com relação à proporção de mortes, entre as duas situações (lavar ou não as mãos) usando um boxplot e estatísticas sumárias que achar relevantes.

```
ggplot(month, aes(y=proportion, x=order_handwash, color=order_handwash))+  
  geom_boxplot() +  
  # Changing the labels on the x-axis  
  scale_x_discrete(labels = c("FALSE" = "Before", "TRUE" = "After")) +  
  # Changing the labels in the legend  
  scale_color_discrete(labels = c("FALSE" = "Before", "TRUE" = "After")) +  
  labs(  
    title = "Distribution of Death Monthly in Clinic 1 ",  
    x = "Date",  
    y = "Proportion of Deaths (%)",  
    colour = "Handwashing Order in Effect"  
  )
```



```
# Calculating the median of the proportion of deaths  
  
median_proportion_death_no_handwash <- median(month$proportion[month$order_handwash == FALSE])  
median_proportion_death_yes_handwash <- median(month$proportion[month$order_handwash == TRUE])
```

```

#Calculating the mode of the distribution

# Function to calculate the mode using Czuber's formula
# Mo = L + [d1 / (d1 + d2)] * h
modal_czuber <- function(x) {
  x <- na.omit(x)                                # remove NA
  h <- hist(x, plot = FALSE)                     # create the classes (bins)
  counts <- h$counts
  i <- which.max(counts)                          # modal class position

  L <- h$breaks[i]                               # lower limit of the modal class
  width <- h$breaks[i + 1] - h$breaks[i]         # class width (h)

  f_prev <- if (i == 1) 0 else counts[i - 1]      # previous class frequency
  f_next <- if (i == length(counts)) 0 else counts[i + 1] # next class frequency

  d1 <- counts[i] - f_prev                       # modal minus previous
  d2 <- counts[i] - f_next                       # modal minus next

  L + (d1 / (d1 + d2)) * width                  # Czuber's formula
}

# Mode with handwashing
mode_proportion_death_yes_handwash <- modal_czuber(month$proportion[month$order_handwash == 1])

# Mode without handwashing
mode_proportion_death_no_handwash <- modal_czuber(month$proportion[month$order_handwash == 0])

```

According to the statistical summary of the monthly proportion of deaths in Clinic 1, the period before the handwashing order had a *mean* = 10.5049983, a *median* = 10.5168832, and a *mode* = 3.3823529. In the period after the order, the *mean* was 2.1093375, the *median* was 2.063595, and the *mode* was 0.7. Although the modes of the two distributions are similar, the mean differs by 8.3956608 percentage points and the median by 8.4532882 percentage points. Both measures differ by approximately 8 percentage points between the periods. Furthermore, the boxplot shows that the distribution before the order was highly dispersed and positively skewed.

## 5.1 Testing Hypothesis

Despite the marked decrease in the proportion of deaths observed after the students began washing their hands, Dr. Semmelweis was met with disbelief. At the time, he did not have the

statistical tools needed to defend his position. In this study, we use a hypothesis test to draw conclusions from the data.

The mean proportion of deaths differs between the two periods (before and after the handwashing order). Can this difference be attributed to handwashing, or could it plausibly arise by chance alone, even if handwashing had no practical effect? To address this question, we adopt as the null hypothesis ( $H_0$ ) that handwashing has no effect on the proportion of deaths, and we assess how likely a difference as large as the one observed would be under that assumption. If this probability is very small, we have evidence against  $H_0$  and in favor of an effect of handwashing.

## References